

VLAN, RIP2, OSPF1, WEB SERVER, DHCP, ACL, NAT, and PAT

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Contents

1	BASIC INSTRUCTIONS	2
2	VLAN Configuration	2
2.1	VLAN Configuration on a Switch	2
2.2	VLAN Configuration on a Router	2
3	RIP Version 2 (RIP v2)	3
3.1	Basic RIP v2 Configuration	3
4	OSPF (Open Shortest Path First)	3
4.1	Basic OSPF Configuration	3
5	Web Server Configuration	4
6	DHCP Server Configuration	5
7	Access Control List (ACL)	5
7.1	Definition	5
7.2	Types of ACLs	5
7.3	Application	5
7.4	Standard ACL Example to Block a VLAN Toward a Server	6
7.5	Extended ACL Example to Filter Traffic Only Toward a Specific Server	6
8	NAT and PAT	7
8.1	Basic NAT	7
8.2	Dynamic NAT with an IP Pool	7
8.3	Dynamic PAT (NAT overload)	8

1 BASIC INSTRUCTIONS

1. **enable** → enters privileged mode (always needed except for show commands, recognized because the prompt changes to "#").
2. **config t** → executed after enable to modify settings (recognized because the prompt shows "(config)").
3. **show running-config** → displays the device configuration (executed outside configuration mode, pressing space shows what is not currently written on the page).
4. **write m** → used to save configurations (must be done before submitting on all router and switch devices, must be executed in privileged mode but not in config mode).

2 VLAN Configuration

2.1 VLAN Configuration on a Switch

1. Config VLAN database → write the name and number of the VLAN and add it
2. For each interface of each device, indicate which VLAN it belongs to (must be of access type)
3. For the interface going toward the router, simply change the mode to "trunk"

2.2 VLAN Configuration on a Router

Activate the router cable.

Then in the CLI:

```
Router(config)# interface fastEthernet 0/0.10
Router(config-if)# encapsulation dot1Q 10
Router(config-if)# ip address ip_gateway netmask (of the VLAN)
Router(config-if)# no shutdown
Router(config-if)# exit (repeat for each VLAN)
```

Explanation:

- interface fastEthernet 0/0.10 → access to the subinterface of VLAN 10.
- encapsulation dot1Q 10 → explains to encapsulate messages from that VLAN using dot1Q.
- ip address ip gateway netmask indicates what gateway IP the VLAN has
- no shutdown instructs to change the interface mode to UP

3 RIP Version 2 (RIP v2)

Executed for each router, each network directly connected to the router must be added (network..)

3.1 Basic RIP v2 Configuration

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.10.0 (base_ip of the adjacent
network)
Router(config-router)# network 192.168.20.0 (base_ip of the adjacent
network)
Router(config-router)# passive-interface fastEthernet 0/0.10 (to be
done only for each VLAN to prevent unnecessary data traffic)
```

Explanation:

- router rip enables the RIP protocol on the router.
- version 2 uses RIP version 2 (supports variable subnet masks).
- network X.X.X.X indicates the directly connected networks that the router must advertise to other routers.
- passive-interface fastEthernet 0/0.10 indicates that packets used to explain neighbors to routers are not forwarded to the VLANs since they do not need them

4 OSPF (Open Shortest Path First)

4.1 Basic OSPF Configuration

```
Router(config)# router ospf 1
Router(config-router)# network 192.168.10.0 0.0.0.255 area 1 (base_ip
of the adjacent network and wildcard mask)
Router(config-router)# network 192.168.20.0 0.0.0.255 area 1 (base_ip
of the adjacent network and wildcard mask)
Router(config-router)# passive-interface fastEthernet 0/0.10 (to be
done only for each VLAN to prevent unnecessary data traffic)
```

Explanation:

- router ospf 1 → enables OSPF with process ID 1.
- network < IP > < wildcard > area < area > → defines which networks to advertise in which OSPF area. The **wildcard mask** is the inverse of the subnet mask: for /24: 255.255.255.0 → wildcard 0.0.0.255, area 1 indicates which area to add the network to
- passive-interface fastEthernet 0/0.10 → indicates that packets used to explain neighbors to routers are not forwarded to the VLANs since they do not need them

5 Web Server Configuration

A **web server** is a host configured to offer the HTTP/HTTPS service. In Packet Tracer it can be easily configured:

1. Add a device of type **Server**.
2. Go to **Services** → **HTTP** and enable http/https.
3. Set the gateway
4. Set the ip
5. Set the netmask

At this point, any PC on the same LAN, opening the browser and typing `http://192.168.10.100`, will display the default page of the server.

To make it accessible from the outside (public network), static NAT must be configured on the router.

In the CLI:

```
! HTTP port mapping from public to internal server
ip nat inside source static tcp 192.168.10.100 80 203.0.113.50 80

! Interface configuration
interface fastEthernet 0/0
ip address 192.168.10.1 255.255.255.0 (server gateway IP)
ip nat inside

interface fastEthernet 0/1
ip address 203.0.113.50 255.255.255.0 (public IP, IP assigned to the
    interface connected toward the outside of the router network)
ip nat outside
```

Thus, anyone accessing `http://203.0.113.50` from the outside will reach the internal web server.

6 DHCP Server Configuration

The **DHCP** service allows automatically assigning IP addresses to network clients. In Packet Tracer, a DHCP server is configured as follows:

1. Add a device of type **Server**.
2. Go to **Services** → **DHCP** and enable the DHCP service.
3. Set default gateway: gateway of the destination VLAN where addresses are to be assigned
4. Start ip address: first IP of the destination VLAN
5. Subnet mask: subnet mask of the destination VLAN
6. Maximum number of users: wildcard - 2
7. Save
8. On the PCs to which the server must assign IPs, set IP to DHCP mode (**Config** → **DHCP**)

Now the router must be configured so that it allows the server to assign the IPs. In the CLI:

```
interface fastEthernet 0/0.10 (interface connected to the switch,  
    specifying the destination VLAN)  
ip helper-address ip_of_the_server
```

7 Access Control List (ACL)

7.1 Definition

An **ACL (Access Control List)** is a list of rules that a router or a Layer 3 switch uses to control traffic passing through interfaces, deciding what is permitted and what is blocked.

WARNING: these rules are checked by the router one after another; if a rule is matched, the subsequent ones are not even checked. For this reason, the last rule is generally to permit everything, since you define first who is denied (or vice versa).

7.2 Types of ACLs

- **Standard:** filter based only on the source IP address. ID number between 1-99
- **Extended:** filter based on source, destination, protocol, and port. ID number between 100-199

7.3 Application

ACLs are applied to the interfaces of a router with the commands:

```
Router(config-if)# ip access-group <number/name> in|out
```

"in" indicates that rules are checked for inbound traffic flowing into that interface, "out" indicates that they are checked for outbound traffic flowing out.

7.4 Standard ACL Example to Block a VLAN Toward a Server

Components:

1. VLAN 1: base ip: 192.168.10.0 netmask: 0.0.0.255
2. VLAN 2: base ip: 192.168.20.0 netmask: 0.0.0.255
3. SERVER: ip: 172.16.0.100

What it does:

1. Deny VLAN_1 from accessing the server
2. Permit everyone else to access everything
3. Assign the ACL to the router interface facing the server on the outbound direction

```
access-list 10 deny 192.168.10.0 0.0.0.255
access-list 10 permit any
interface fastEthernet 0/1
ip access-group 10 out
```

7.5 Extended ACL Example to Filter Traffic Only Toward a Specific Server

Components:

1. VLAN 1: base ip: 192.168.10.0 netmask: 0.0.0.255
2. VLAN 2: base ip: 192.168.20.0 netmask: 0.0.0.255
3. SERVER: ip: 172.16.0.100

What it does:

1. Permit VLAN 1 to access the server
2. Deny everyone else from accessing the server
3. Permit everyone to access everything else
4. Assign the ACL to the router on the inbound interface

```
access-list 110 permit ip 192.168.10.0 0.0.0.255 host 172.16.0.100
access-list 110 deny ip any host 172.16.0.100
access-list 110 permit ip any any
interface fastEthernet 0/0
ip access-group 110 in
```

8 NAT and PAT

8.1 Basic NAT

Allows translating all internal addresses into the public address of the router and vice versa.

Components:

1. fastEthernet 0/0: interface connecting the internal network/VLAN to the router
2. fastEthernet 0/1: interface connecting the external network to the router

What it does:

1. Declare that 0/0 is the internal network
2. Declare that 0/1 is the external network

```
interface fastEthernet 0/0
ip nat inside
exit
interface fastEthernet 0/1
ip nat outside
exit
```

8.2 Dynamic NAT with an IP Pool

If instead we want the internal IPs to be translated into multiple public IPs, an IP pool is used, which consists of defining with which range of public IPs the internal ones must be translated.

Components:

1. MIO_POOL: pool name
2. Range of public addresses: 203.0.113.10-203.0.113.20, netmask: 255.255.255.0
3. VLAN-1: base.ip: 192.168.10.0, netmask: 0.0.0.255
4. SERVER: ip: 172.16.0.100

What it does:

1. Set the pool name with its range of external addresses
2. Specify which is the base address of the VLAN/individual host to be translated
3. Command to use the IPs in the pool to translate internal addresses

```
ip nat pool MIO_POOL 203.0.113.10 203.0.113.20 netmask 255.255.255.0
access-list 1 permit 192.168.10.0 0.0.0.255
ip nat inside source list 1 pool MIO_POOL
```

8.3 Dynamic PAT (NAT overload)

Configured so that, having a single public IP, multiple hosts can exit using the same IP and not one at a time, because their port number is used to differentiate and recognize them.

Components:

1. VLAN 1: base ip: 192.168.10.0 netmask: 0.0.0.255

What it does:

1. Set up as for normal NAT (inside/outside interfaces)
2. Specify which is the base address of the VLAN/individual host to be translated via an ACL
3. Instruct to use the IP of interface 0/1 (router public interface) to translate addresses, "overload" indicates that multiple hosts can exit with that IP simultaneously and not one at a time

```
interface fastEthernet 0/0
ip nat inside
interface fastEthernet 0/1
ip nat outside
access-list 1 permit 192.168.10.0 0.0.0.255
ip nat inside source list 1 interface fastEthernet 0/1 overload
```